

MUHAMMAD RAFIE HABIBI FAUZI

Tangerang | mrafiehabibi01@gmail.com | linkedin.com/in/rafiehabibi/

Computer Engineering student at University of Brawijaya with hands-on experience in IoT device deployment and embedded systems development, including real-world project experience with ESP32, Arduino, and sensor integration. CCNA and MTCNA certified, with a strong foundation in network infrastructure and Linux system administration supporting IoT connectivity and device management.

EDUCATION

Brawijaya University, Malang

Aug 2023 – Present

Bachelor Degree of Computer Engineering

GPA 3.82 of 4.00

WORK EXPERIENCE

Network Operation Center(NOC)

Feb 2026 – Present

PT Nale System Integrator, Malang

- Monitored network uptime, performance, and security across 2+ client sites.
- Identified and resolved Tier 1 network incidents and connectivity issues.
- Analyzed security threats and compiled monthly threat intelligence reports month.

Staff IT Infrastructure

Jan 2026 – Feb 2026

PT Cipta Harmoni Lestari, Tangerang

- Deployed and configured of IT network and IoT systems for 3 home clients, 3 office buildings, and 1 cluster environment.
- Configure IT systems and network equipment.

Microcontroller Practicum Assistant

Aug 2025 – Dec 2025

Brawijaya University, Malang

- Assisted in teaching and supervising 30+ students in Microcontrollers practicum course.
- Guided students in programming microcontrollers using C and C++, including GPIO, interrupts, and peripheral interfacing.
- Prepared experiment modules, demonstrated procedures, and troubleshoot hardware and firmware issues during lab sessions.

Advanced Electronic Circuits Practicum Assistant

Feb 2025 – Jun 2025

Brawijaya University, Malang

- Assisted in teaching and supervising 30+ students in Advanced Electronic Circuits practicum course.
- Guided students in analyzing and building circuits involving op-amps, filters, amplifiers, and feedback systems.
- Prepared and set up lab experiments, demonstrated measurement procedures, and ensured safe handling of equipment.

Digital System Practicum Assistant

Aug 2024 – Des 2024

Brawijaya University, Malang

- Assisted in teaching and supervising 30+ students in Digital Systems practicum course.
- Guided students in understanding logic gates, combinational and sequential circuits, and FPGA/HDL fundamentals.
- Prepared experiment setups, demonstrated circuit procedures, and troubleshoot digital circuit issues during lab sessions.

Assistant of Robotics and Embedded System Laboratory

Feb 2024 – Present

Brawijaya University, Malang

- Manage laboratory operations by overseeing inventory, maintaining and troubleshooting equipment, and supporting practicum activities.

SKILLS

Hard Skill

- Tools Networking & Infrastructure : MikroTik, Cisco, UniFi, PNETLab, Cisco Packet Tracer, Linux
- Tools IoT : Arduino, ESP32, MQTT, 3D Design, LoRA, Node-RED, Blynk, OpenCV.
- Virtualization : Proxmox, VMware, Docker, VirtualBox.
- Tools Data Visualisation : Prometheus, Grafana.
- Programming & Automation : Python, C++, C, SQL.

Languages

- English (Professional Working Proficiency)
- Bahasa Indonesia (Native)

LEADERSHIP & ORGANIZATION ACTIVITIES

ROVER(Robotics Value Empowerment), Malang

May 2024

Head of Event

- As a leader, I was responsible for leading the committee in organizing the event and ensuring its successful execution.

ROBOTIK, Malang

May 2023

Staff of Competition Division

- Oversee team performance, timeline achievement, and schedule the teams coordination meeting.

PROJECTS

- **Smart Plant Care** — Built an ESP32-based smart plant monitoring system with real-time sensor data (soil moisture, temperature, light) transmitted to Blynk dashboard via Wi-Fi with automated water pump control.
- **Autonomous Car** — Developed an Arduino-based autonomous vehicle capable of GPS-guided navigation with real-time PID heading correction using a compass sensor (QMC5883) and obstacle detection via ultrasonic sensor (HC-SR04)
- **ML Predict Dissolved Oxygen in Water** — Developed a Multiple Linear Regression model to predict dissolved oxygen levels in water based on environmental sensor parameters (pH, temperature, turbidity, conductivity) with 71% accuracy.
- **Obstacle Detection App with Optical Flow** — Developed a real-time obstacle detection using Dense Optical Flow (Farneback algorithm) with multi-metric analysis (magnitude, temporal integration, looming detection) and 3-zone spatial awareness via OpenCV.